UT : 2 QB

Module : 4 : Counting

Questions :

1) State the pigeonhole principle and show that if any five numbers from 1 to 8 are chosen, then two of them will add to 9.

2) Find the Number of integers between 1 to 500 that are not divisible by 5, 6, or 8 ?

3) How many bit strings of Length 7 either begin with 2 zeroes or end with 3 ones ?

4) A Large software development company employs 100 computer programmers. Of them, 45 are proefficient in Java, 30 in C#, 20 in Python, Six in C# and Java, one in Java and Python, Five in C# and Python, and just one programmer is proefficient in all three languages above. Determine the No. of computer programmers that are not proefficient in any of these three languages.

5) Determine the Sequence b_n whose recurrence relation is

$$b_n = 2b_{n-1} + 1 \text{ with initial Condition } \boxed{b_1 = 7}$$

Module : 5 : Algebraic Structures.

Questions :

1) Show that $A = \{0, 3, 6, 9, 12\}$ is a ring w.r.t. the operation of addition and multiplication modulo 15.

2) What is an Abelian Group ? Let $G = \{1, 2, 3, 4, 5, 6, 7\}$

i) Prepare the Composition table w.r.t. the operation of multiplication modulo 8.

ii) Check whether it is an Abelian group ? Justify your Answer.

3) Define minimum hamming distance. Find the code words generated by the parity check matrix H given below :

$$H = \begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

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4) Let $A = \{0, 1, 2, 3, 4, 5\}$

i.) Explain the term group.

ii.) Prepare the composition table for the above set w.r.t. the operation of addition modulo 6

iii) Determine whether it is a group.

iv) Whether elements of set A are invertible? If yes, then find the inverses of these elements.

v.) Determine whether it is a cyclic group.

5) Consider (3,8) an encoding function $e: B^3 \rightarrow B^8$ defined as:

$e(000) = 00000000$

$e(001) = 10111000$

$e(010) = 00101101$

$e(011) = 10010101$

$e(100) = 10100100$

$e(101) = 10001001$

$e(110) = 00011100$

$e(111) = 00110001$

How many errors can 'e' detect and correct?

6) If the addition and multiplication modulo 10 is defined on a set of integers $A = \{0, 2, 4, 6, 8\}$. Then determine whether this algebraic system is a ring.

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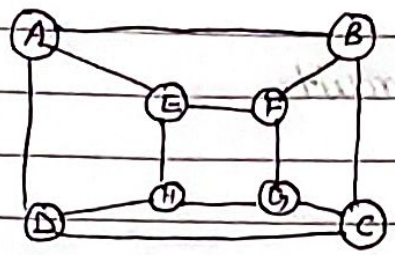
Module: 6 : Graph Theory

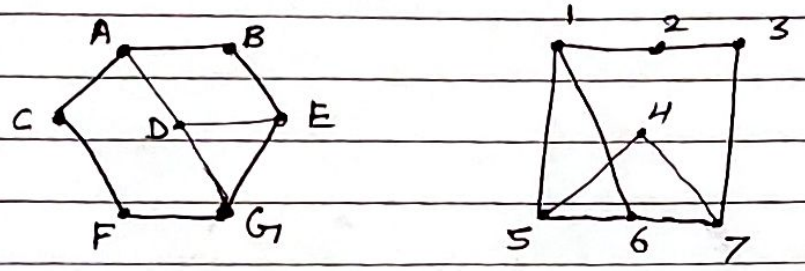
Questions:

1) Define the following terms
   i. Planer graph
   ii. Cut vertex
   iii. Chain
   iv. Monoid
   v. Group

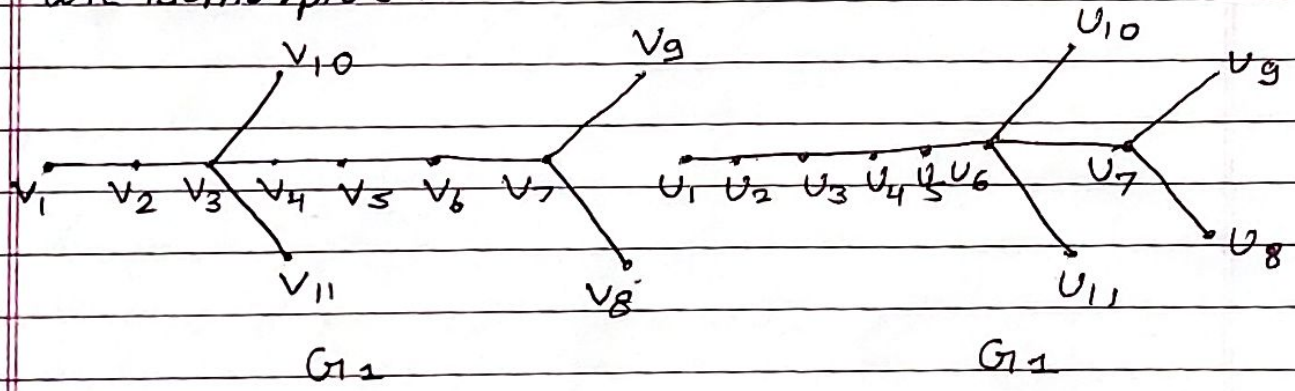
2) Determine the following graph is Eulerian or Hamiltonian or both. Justify your answer.



3) Check whether the following graphs are Isomorphic or not? Justify.

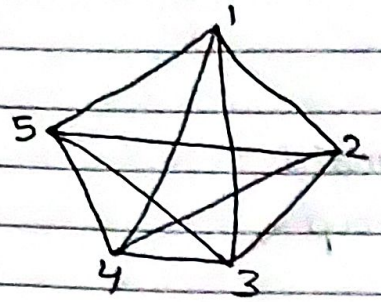


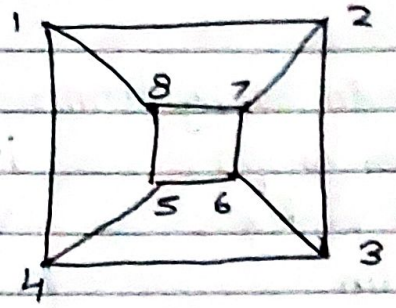
4) What are the necessary conditions for the isomorphism between 2 graphs? Determine whether following 2 graphs are isomorphic.



Q.5) a) Define the terms Euler path and a circuit

b) Determine whether following graphs have Euler path or a circuit.





6) a) Explain the following terms with suitable example

i.) Hamiltonian path and circuit
ii.) Bipartite graph
iii.) Adjacency matrix
iv) Equivalence relation
v) Cartesian product.